

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 50%. We keep your highest score.

Next item →

1. How does dropout regularization improve the generalization ability of a neural network?

1 / 1 point

- ☐ By increasing the training data
- ☒ By forcing the network to learn redundant representations
- ☐ By ensuring each neuron is used only once during training
- ☐ By reducing the complexity of the model
- ✔ Correct

Correct! Dropout regularization improves generalization by ensuring the network learns redundant representations, reducing overfitting.

2. Identify the benefits of applying 'same' padding in convolutional layers.

1 / 1 point

- ☐ Improves the convergence rate during training
- ☒ Facilitates the use of deeper networks
- ✔ Correct

Well done! 'Same' padding can facilitate the use of deeper networks by preserving spatial dimensions.
- ☒ Maintains spatial dimensions of the input
- ✔ Correct

Correct! 'Same' padding maintains the spatial dimensions of the input.
- ☐ Increases the computational efficiency of the network
- ☒ Simplifies the architecture design
- ✔ Correct

Right! 'Same' padding can simplify the architecture design by maintaining consistent dimensions.

3. What is the typical effect of using 'same' padding in convolutional layers?

1 / 1 point

- ☒ The output feature map has the same spatial dimensions as the input
- ☐ The output feature map has increased spatial dimensions compared to the input
- ☐ The output feature map has random spatial dimensions
- ☐ The output feature map has reduced spatial dimensions compared to the input
- ✔ Correct

Correct! 'Same' padding ensures that the output feature map maintains the same spatial dimensions as the input.

4. Select the transformations that can be part of image data augmentation.

1 / 1 point

- ☐ Color inversion
- ☒ Rotation
- ✔ Correct

Correct! Rotation changes the angle of the image, adding variety.
- ☒ Height shift
- ✔ Correct

Correct! Height shift can be used to move the image vertically.
- ☒ Horizontal flip
- ✔ Correct

Correct! Horizontal flip is used to mirror the image, enhancing diversity.
- ☒ Rescaling
- ✔ Correct

Correct! Rescaling changes the size of the image, making it a key augmentation technique.
- ☐ Random cropping

5. Which of the following techniques is used to change the angle of an image?

1 / 1 point

- ☐ Height shift
- ☐ Rescaling
- ☒ Rotation
- ☐ Width shift
- ✔ Correct

Correct! Rotation changes the angle of the image.